

Municipal Waste Minimisation and Circular Economy Planning

OEG Graduate Programme Portfolio Case Study

Simulated role: Graduate Environmental & Infrastructure Analyst

Disclosure: Mavhungu Local Municipality and all operational data are fictional/synthetic. This independent portfolio project is not presented as professional experience for a real municipality.

1. Executive Summary

This case study examines how a fictional municipality could improve waste management through data analysis, environmental risk assessment and constrained-budget intervention planning. The analysis considers waste generation, material recovery, landfill dependence, collection performance and illegal dumping. A transparent prioritisation model is used to examine how an illustrative R15 million budget could be allocated across interventions.

2. Problem Definition

The municipality faces increasing waste volumes, limited material recovery, landfill dependence and uneven service performance. The analytical objective is to identify measurable performance gaps and develop an evidence-based intervention portfolio rather than simply present a descriptive dashboard.

3. Methodology

Monthly synthetic observations were generated for six service zones from January 2025 to August 2026. Calculated indicators include recovery rate, landfill tonnage, cost per tonne, fuel intensity, illegal-dumping incidence and an environmental risk score. The risk score combines illegal dumping, landfill burden, missed collections and recovery performance using explicit weights. Intervention prioritisation uses benefit score per rand subject to an illustrative R15 million budget.

4. Key Indicators

Zone	Waste (t)	Recovery %	Landfilled (t)	Dumping	Risk
Central	859	15.9%	723	208	35.9
East	669	10.4%	599	262	44.8
North	1,120	12.9%	976	142	52.4
Rural	547	8.4%	501	337	75.0
South	975	12.8%	850	164	48.6
West	752	18.0%	617	100	15.4

5. Intervention Prioritisation

The illustrative budget is R15 million. The selected portfolio costs R13,400,000 and has an estimated annual landfill-diversion potential of 14,000 tonnes under the stated synthetic assumptions. The model is deliberately transparent and should be treated as a screening/prioritisation exercise rather than a final investment decision.

Intervention	Cost	Expected diversion (t)	Selected
Collection route optimisation	R1,000,000	1,800	Yes
Community awareness programme	R900,000	900	Yes
Material recovery partnership	R2,500,000	2,600	Yes
Recycling drop-off centres	R3,000,000	3,100	Yes
Separation at source	R4,000,000	4,200	Yes

Intervention	Cost	Expected diversion (t)	Selected
Illegal dumping prevention	R2,000,000	1,400	Yes
Organic waste recovery	R5,000,000	5,200	No

6. Environmental and Regulatory Context

South Africa's National Environmental Management: Waste Act 59 of 2008 provides the legislative framework for waste management, including planning, norms and standards, licensing and compliance. The National Waste Management Strategy 2020 places circular economy principles at the centre of the national waste strategy and identifies waste minimisation, effective and sustainable waste services, and compliance/enforcement/awareness as strategic pillars.

This case study does not perform a legal compliance determination. Regulatory references are used to frame the analytical context only.

7. Recommendations

1. Prioritise interventions using measurable environmental and service indicators rather than expenditure alone.
2. Improve source separation and material recovery where the model indicates a strong diversion opportunity.
3. Target illegal-dumping hotspots with service, infrastructure and community interventions rather than relying on a single measure.
4. Track cost per tonne, recovery rate, landfill tonnage, missed collections and dumping incidents as recurring management KPIs.
5. Validate all assumptions with municipal operational data before implementation.

8. Limitations

All municipal observations, costs, diversion estimates and risk scores are synthetic. Real projects would require verified waste quantities, route data, facility capacity, regulatory requirements, procurement information, stakeholder consultation and site-specific environmental assessments.

9. Skills Demonstrated

Environmental management; sustainability analysis; quantitative analysis; Excel modelling; infrastructure planning; risk assessment; optimisation; technical reporting; and decision support.

Sources

South African Government, National Environmental Management: Waste Act 59 of 2008. Department of Forestry, Fisheries and the Environment, National Waste Management Strategy 2020.